
❑ Kubernetes Interview Questions – Extracted (Questions Only)

Ready to use for self-practice or mock interviews

1❑ Kubernetes Fundamentals

1. What is Kubernetes and why is it used?
 2. What problems does Kubernetes solve that Docker alone cannot?
 3. Explain Kubernetes architecture at a high level.
 4. What is a cluster in Kubernetes?
 5. Difference between containerization vs orchestration?
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2❑ Kubernetes Architecture (Control Plane & Nodes)

6. Explain the role of:
 - API Server
 - etcd
 - Scheduler
 - Controller Manager
 7. What happens if etcd goes down?
 8. What is a node?
 9. What components run on a worker node?
 10. What is kubelet?
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3❑ Pods & Containers

11. What is a Pod?

12. Why is Pod the smallest deployable unit?

13. Can a Pod have multiple containers? When would you do that?

14. What happens when a Pod dies?

15. Difference between:

- Pod vs Deployment
 - Pod vs ReplicaSet
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4 ☐ Workloads

16. What is a Deployment?

17. Difference between:

- Deployment vs StatefulSet
- Deployment vs DaemonSet
- DaemonSet vs Job
- Job vs CronJob

18. What is a ReplicaSet?

19. How does Kubernetes maintain desired state?

5 ☐ Services & Networking

20. What is a Service?

21. Explain Service types:

- ClusterIP
- NodePort
- LoadBalancer
- ExternalName

22. Difference between ClusterIP vs NodePort

23. How does Pod-to-Pod communication work?

24. What is kube-proxy?

25. How does Kubernetes DNS work?

6 Ingress & Traffic Management

26. What is Ingress?

27. Difference between Service vs Ingress

28. What is an Ingress Controller?

29. Name popular Ingress controllers.

30. How do you expose multiple services with one LoadBalancer?

7 Config & Secrets

31. What is a ConfigMap?

32. What is a Secret?

33. Difference between ConfigMap vs Secret

34. How do you mount secrets into pods?

35. Why should secrets not be stored in Git?

8 Storage

36. What is a Volume?

37. Difference between emptyDir vs hostPath

38. What is a PersistentVolume (PV)?

39. What is a PersistentVolumeClaim (PVC)?

40. Difference between PV vs PVC

41. What is a StorageClass?

9 ☐ Scheduling & Scaling

- 42. What is the Kubernetes scheduler?
- 43. What is HPA (Horizontal Pod Autoscaler)?
- 44. What metrics does HPA use?
- 45. Difference between HPA vs VPA
- 46. What is Cluster Autoscaler?

☐ Security & Access Control

- 47. What is RBAC?
- 48. Difference between Role vs ClusterRole
- 49. What is a ServiceAccount?
- 50. What is PodSecurity (PSA)?
- 51. How do you secure a Kubernetes cluster?

☐ Scenario-Based Questions

- 52. Your pod is in CrashLoopBackOff. How do you debug it?
- 53. Pod is running but application is not accessible externally. How do you debug?
- 54. How do you deploy a new version without downtime?
- 55. Pods are stuck in Pending state. Why?
- 56. ConfigMap updated but pods didn't pick up changes — why?
- 57. How do you manage secrets securely in production?

- 58. A namespace consumes too many resources. How do you control it?**
 - 59. A node goes down. What happens to pods?**
 - 60. How do you upgrade a Kubernetes cluster safely?**
 - 61. How do you manage Dev, QA, and Prod environments in Kubernetes?**
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☐ **Advanced Kubernetes Questions**

- 62. What is etcd quorum?**
 - 63. How does Kubernetes achieve self-healing?**
 - 64. What are Custom Resource Definitions (CRDs)?**
 - 65. What is an Operator?**
 - 66. What is a Service Mesh?**
 - 67. How is Kubernetes networking different from Docker networking?**
 - 68. How do you debug DNS issues in Kubernetes?**
 - 69. What is an Admission Controller?**
 - 70. What is a Pod Disruption Budget (PDB)?**
 - 71. What is OOS (Out of Sync) in Kubernetes?**
 - 72. What is QoS in Kubernetes?**
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☐ **Extra: DNS & Headless Services**

- 73. What is Kubernetes DNS?**
- 74. What is a Headless Service?**
- 75. What DNS records does Kubernetes use (A, AAAA, CNAME)?**